

Automating Browsers with Ruby and Watir-webdriver

Lightweight and simple to use, Watir is a browser driver that is great for automating modern browsers. It is built on Ruby but it also works well with other testing frameworks supported by Ruby. In this article, the author describes the installation of Watir and its use in retrieving the results of the CBSE Board exams.



If necessity is the mother of invention, then boredom is the grandmother of automation. Automation refers to the process of performing a task without any attention or input required from the person performing that task. I am referring to tasks that are quite trivial, while at the same time boring, like getting results from a website based on a query or a user selection. Most of the time, things like this can easily be done by Web scraping or vendor APIs, but we can also instruct browsers to do these systematically. In this article, we will use Ruby as the commanding force and Watir-webdriver (Watir stands for Web automation testing in Ruby) as the bridge between Ruby and the browser. Although Watir is mainly used for Web application testing, we can also use it to create useful scripts. And here's a small note: after you finish this article, you may want to get deeper into Watir, and while exploring the Web, you will usually see both Watir and Watir-webdriver; don't get confused. The difference lies in the browser that is driven. Watir only works/drives with Internet Explorer on Windows, while Watir-webdriver drives Firefox and Chrome on Windows, Mac and Linux.

Installation

Let's get started. Since this set-up requires the use of Ruby, the first obvious step is to check for Ruby. If Ruby is installed on your system then your development environment is already prepared (which is no easy task); otherwise, install Ruby. Depending on your operating system and distribution, the procedure's steps may differ. When in doubt, refer to the installation documentation available at <https://www.ruby-lang.org/en/documentation/installation/>. But whichever way you choose, always remember to use version management tools

like *rbenv*, *rvm* or *pik*. For Linux and Mac, the best way to install Ruby is through *rbenv* or *rvm* and for Windows, it is *pik* (no longer maintained) or *Uru* (though, I would advise you to stick with UNIX systems, as managing Ruby in Windows is difficult compared to Mac and Linux). Check the 'References' section for links of detailed tutorials. Since we have prepared our development environment, we can proceed to install libraries for Watir or 'Gems' as we call them in the Ruby world. Open the terminal or the equivalent to install Gems.

To install Watir-webdriver, use the following command:

```
gem install watir-webdriver
```

If the following commands throw errors due to lack of privileges (usually encountered on Ubuntu), prefix the command with *sudo*:

```
sudo gem install watir-webdriver # Use this if previous command doesn't work
```

To install Watir, issue the following command:

```
gem install watir # Prefix with sudo if necessary
```

Fire up *irb* or *pry* (whichever suits you) to load the Watir-webdriver gem:

```
require 'watir-webdriver'
```

Now, to invoke a new browser, use the following command: